Together First, Apart Later? When Optimizers Start to Learn Different Functions

Initial checkpoint, MIT 6.7910, Fall 2026

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Summary. We train one transformer from one initialization on one sequence of mini-batches with AdamW and with Muon (Jordan et al., 2024). (1) At equal training loss their predictions are as close as those of two data orders of one optimizer; the loss barrier between their endpoints exceeds the data-order ones on one of three datasets. (2) Along the way, metrics of their weights and updates differ far beyond the data-order spread, metrics of their predictions do not: Muon's updates and weights have flat spectra, AdamW's a few dominant directions, fewer active units and a flatter loss. (3) Swapping the two mid-training leaves the function within the floor of both, while the weights keep part of the parent's geometry.

Setup. The main setting is TinyStories (Eldan and Li, 2023) with a 14M-parameter nanoGPT (Karpathy, 2022) (6 layers, width 384, 10,000 steps of 128×256 tokens, cosine decay); a 30M model on TinyStories and a character-level model on Shakespeare are checks (Appendix A). Muon orthogonalizes the update of each block matrix and leaves embeddings, head and gains to an auxiliary AdamW; each optimizer uses the best rate of a five-rate sweep. All runs share one initialization; a data order fixes the mini-batch at every step (five per optimizer). The distance between two runs is the mean Jensen–Shannon divergence (JS, bits) of their next-token distributions on 2,048 fixed held-out positions. We compare runs at equal *training loss* (on 64 fixed training sequences), so that speed does not count, and against a *floor*: the distance between two data orders of one optimizer (the larger of the two means). In all, 123 runs and 31 GPU-hours, with 18 more running (Appendix D).

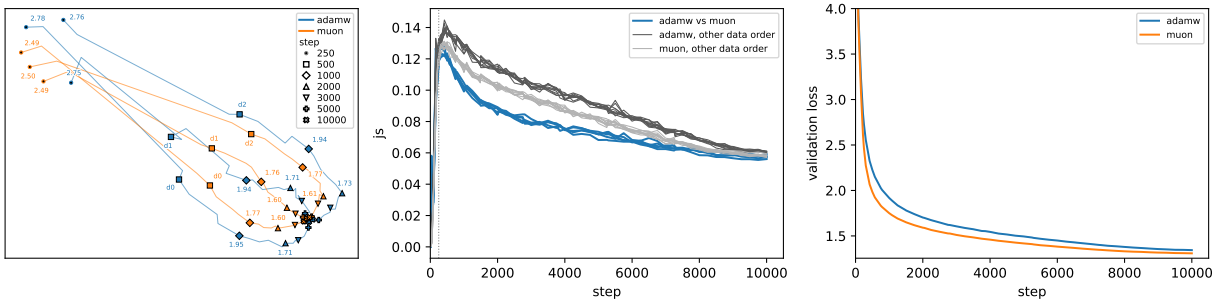


Figure 1: TinyStories-10M, AdamW (blue) and Muon (orange). Left: data orders d0–d2 from step 250 on, embedded with densMAP from the distances $\sqrt{\text{JS}}$ between all logged steps (points at least 0.28 apart); numbers: training loss at steps 250, 1000, 2000. Middle: JS between AdamW and Muon on the same data order at equal steps (blue, five orders) and between two data orders of one optimizer (grey); dotted: step 250. Right: validation loss, mean and range of five orders.

Step 1: how the trajectories differ (Fig. 1). After warmup, AdamW and Muon on the same data order stay close: each such pair are nearest neighbours in the embedding, and at equal steps their distance is below that between two data orders of one optimizer. Muon is ahead (validation loss 1.75 against 1.92 at step 1000, means of five data orders; it reaches AdamW's final 1.344 by step 7500). Compared at equal training loss instead, the distance between AdamW and Muon on the same data order stays at the data-order floor after warmup: 0.95–1.38 floors below loss 4.85 (median 1.02), and 0.99 at the lowest loss all runs reach (TinyStories-30M and Shakespeare: 1.02). In function space both reach the same endpoint, up to the data-order noise.

In weight space they do not. The training loss on the segment between two final weights rises above the line between its end values by a *barrier* (maximum over 11 points; Frankle et al., 2020): 3.5–4.1 nats between AdamW and Muon on TinyStories-10M, against 1.2–1.9 between two data orders of AdamW and 1.6–1.8 of Muon. Only there is the change of optimizer clearly the larger: on TinyStories-30M its mean (4.64) lies between Muon's data-order mean (1.78) and AdamW's (5.05), and on Shakespeare (2.42) between AdamW's (0.14) and Muon's (4.34).

Step 2: what differs along the way. We measure 31 metrics of the models along training up to their endpoints, at 34 checkpoints per run, grouped as weights, one-step updates, curvature, activations and predictions (Appendix B). As for the functions, we compare them at equal training loss and score each gap between two optimizers on the same data order against the gap between two data orders of one optimizer, three data orders each (Fig. 2 and Table 4, which also include SGD with momentum for reference).

Of the metrics we track, those of the weight spectra (spectral norm, stable and effective rank, singular-value spread) separate AdamW from Muon far beyond the data-order spread (|score| 74–263), while those of the predictions (entropy, confidence, calibration, accuracy, loss on frequent and rare targets) do not (|score| ≤ 1.1), nor does the distance moved (2.18 and $2.155\|\theta_0\|$ from the initialization; values at loss 1.81, scores in brackets). Muon's weights have the flatter

spectra: stable rank 86 against 33 (−147), $\sigma_1/\sigma_{\text{median}}$ 2.7 against 8.0 (105). AdamW's model is sparser and louder, on a flatter loss: 12% of MLP units active against 22% (−62), residual RMS 4.7 against 3.4 (28), attention entropy 2.5 against 2.8 nats (−18), tr H 1400 against 2770 (−23), $\lambda_{\max}$ 6.2 against 14 (−10). The high rank of Muon's updates (220 against 17, −328) holds by construction, as Muon orthogonalizes them. SGD is the outlier in how it moves: $0.49\|\theta_0\|$ from the initialization (50 and 58 against AdamW and Muon), steps of 0.06% of $\|\theta\|$ against 0.66% (67, 132), better aligned with their batch gradient (cosine 0.45 against 0.24 and 0.22; −17, −18), on a sharper loss (tr H 25,500; −22, −20).

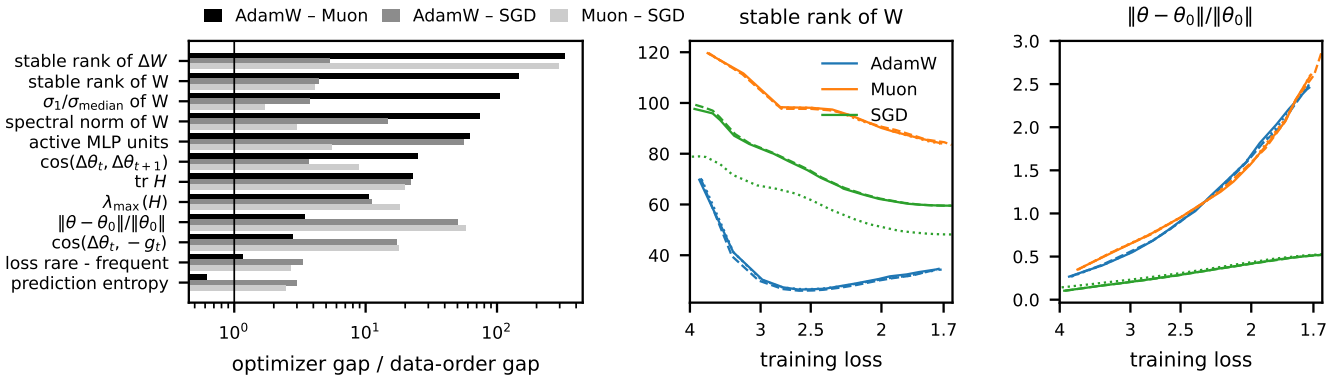


Figure 2: TinyStories-10M, three data orders per optimizer. Left: per metric and pair of optimizers, the mean gap on the same data order over the larger mean gap between two data orders of one optimizer, at equal training loss, median over losses 1.66–3 (selection; all in Table 4). Δ: change over one step; g_t: gradient on the step's batch; H: Hessian on 32 fixed training sequences. Middle, right: two metrics (line style: data order).

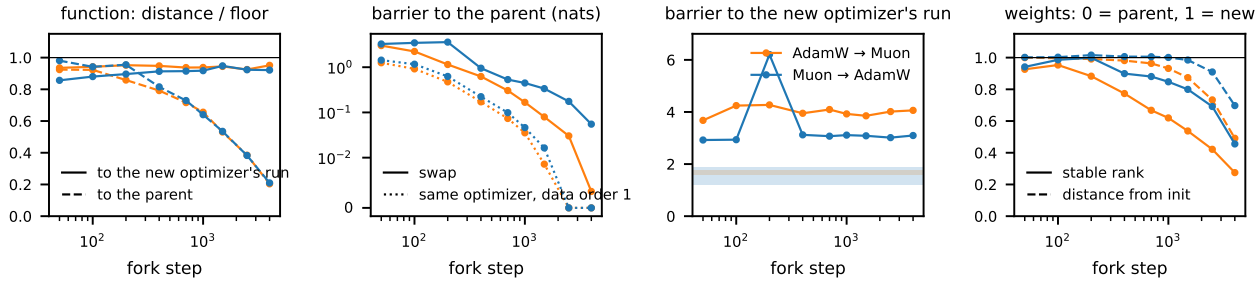


Figure 3: Swaps between AdamW and Muon on TinyStories-10M (data order 0), final weights of each fork against the fork step. Left: JS to the new optimizer's pure run and to the parent, each over that run's data-order floor. Second: barrier to the parent, for the swap and for the parent's optimizer continued on data order 1. Third: barrier to the new optimizer's pure run; bands: its data-order barriers. Right: stable rank and distance from the initialization, as a fraction of the way from the parent's value (0) to the new optimizer's (1).

Step 3: swaps (Fig. 3). We switch between AdamW and Muon at nine fork steps from 50 to 4000, with a warm optimizer state and the parent's batches (Appendix C). Every fork ends within the data-order floor of the new optimizer's pure run (0.86–0.95 floors) and, increasingly with the fork step, of the parent's (0.92–0.98 at step 50, 0.20–0.21 at step 4000): the function cannot tell which optimizer decided it. The weights can. A fork stays linearly connected to its parent (barrier below 0.05 nats) only from a later fork step than a change of data order: from AdamW at step 2500 (0.03) against 1000 (0.035), from Muon not even at step 4000 (0.055) against 1000 (0.046), and none ends in the basin of the new optimizer's run (third panel). Switching at step 4000 moves the stable rank only 0.27 (into Muon) and 0.46 (into AdamW) of the way to the new optimizer's value, and the distance from the initialization 0.49 and 0.70.

Next. (i) TinyStories-30M and Shakespeare; (ii) does the geometry gap explain the barriers? (iii) talk (Dec 1–4), paper, code, Colab (Dec 5–11). Code: `https://github.com/qqgjyx/MIT-6.7910`.

References

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

Frankle, J., G. K. Dziugaite, D. M. Roy, and M. Carbin (2020). "Linear Mode Connectivity and the Lottery Ticket Hypothesis". In: *ICML*.

Jordan, K., Y. Jin, V. Boza, et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*. Blog post. URL: https://kellerjordan.github.io/posts/muon/.

Karpathy, A. (2022). *nanoGPT*. Software. URL: https://github.com/karpathy/nanoGPT.

A Hyperparameters

Table 1: TinyStories-10M, shared by the AdamW and Muon runs. Values from the repository's configuration files and code.

Runs	AdamW and Muon × data orders d0–d4 (10)
Model	nanoGPT, pre-LayerNorm blocks
Layers, heads, width	6, 6, 384
MLP	GELU, width 1536
Biases, dropout	none, none
Output head	untied
Attention projections	separate query, key, value
Context	256 tokens
Parameters	13.9M (10.6M without token and position embeddings and head)
Data	TinyStories (GPT-4 version)
Tokenizer	byte-level BPE, 4096 tokens, trained on the training split
Batch	128 windows × 256 tokens
Window offsets	from the training split; generator seeded with (data order, step), same for every optimizer
Steps	10,000
Tokens	328M (about 0.6 epochs)
Learning rate	$\eta\,\varphi(t)$ at step t, every parameter group
Warmup	linear, $\varphi(t) = (t+1)/200$ for $t < 200$
Decay	cosine to $\varphi = 0.1$ at step 10,000
Gradient clipping	none
Initialization seed	0, all runs
Training precision	bfloat16 autocast
Prediction, Hessian precision	float32, no autocast (TF32 matrix products allowed)
Logged predictions	next-token distributions: 128 fixed validation sequences × 16 evenly spaced positions (2,048), as float16 log-probabilities
Loss for matching runs	64 fixed training sequences, all positions
Both logged at	every 250th step + 30 log-spaced steps in $[1, 10{,}000]$
Hessian	top eigenvalue, 20 Lanczos iterations, 32 fixed training sequences; AdamW also of $P^{-1/2}HP^{-1/2}$
Measured at	every 500th step + 30 log-spaced steps
Loss barrier	64 fixed training sequences, 11 evenly spaced points of the segment between the final weights

Table 2: Per optimizer, TinyStories-10M runs.

	AdamW	Muon
η	$2 \cdot 10^{-3}$	0.01
η choice	best final validation loss of a 5-rate sweep on d0	as AdamW
Update	bias-corrected moments	Nesterov momentum; 5 quintic Newton–Schulz steps in bfloat16
Momentum	$\beta = (0.9, 0.95)$	$\beta = 0.95$
ϵ	10^{-8}	–
Update scale	–	$\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Weight decay	0.1, decoupled; none on LayerNorm gains	none
Parameters	all	query, key, value, attention-output and both MLP matrices of every block
Aux. AdamW	–	embeddings, head, LayerNorm gains
Aux. η, β, ϵ	–	$2 \cdot 10^{-3}$, $(0.9, 0.95)$, 10^{-8}
Aux. weight decay	–	none

Table 3: Shakespeare-char and TinyStories-30M: rows that differ from TinyStories-10M; all else as in Tables 1 and 2. "Aux.": Muon's auxiliary AdamW.

	Shakespeare-char	TinyStories-30M
Runs	AdamW, Muon × d0–d4 (10)	AdamW, Muon × d0–d4 (10)
Data	Shakespeare, nanoGPT's 90/10 split	TinyStories
Vocabulary	65 characters	4096 (BPE)
Layers, heads, width	6, 6, 384	8, 8, 512
Parameters (without embeddings, head)	10.8M (10.6M)	29.5M (25.2M)
Batch	64×256	128×256
Steps	1,500	20,000
Tokens	24.6M (about 24 passes over the 1M training characters)	655M (about 1.2 epochs)
Warmup steps	100	400
Cosine decay to step	1,500	20,000
η_{AdamW}	$3 \cdot 10^{-4}$	$2 \cdot 10^{-3}$
η_{Muon} (aux.)	0.04 (10^{-3})	0.01 ($1.5 \cdot 10^{-3}$)
η chosen by	minimum validation loss over training (runs overfit)	best final validation loss
AdamW β	$(0.9, 0.99)$, as nanoGPT	$(0.9, 0.95)$
Predictions	all 256 positions of 64 validation sequences (16,384)	16 positions of 128 sequences (2,048)
Predictions at	every 50th + 40 log-spaced	every 500th + 30 log-spaced

B Metrics

The runs of Step 2 save checkpoints every 1000 steps, at 12 log-spaced steps, and at three consecutive steps $t, t+1, t+2$ for 24 values of t from 25 to 9998; the metrics are measured at 34 of them per run, the swaps of Step 3 at their final weights. *Weights*: over all parameters, $\|\theta\|_2$, $\|\theta - \theta_0\|/\|\theta_0\|$ and $\|\theta\|_\infty/\|\theta\|_2$; means over the 36 matrices W of the blocks of the stable rank $\|W\|_F^2/\|W\|_2^2$, the effective rank $\exp H(\sigma/\sum\sigma)$ of the singular values σ, the spectral norm and $\sigma_1/\sigma_{\text{median}}$. *Updates* $\Delta\theta_t = \theta_{t+1} - \theta_t$: $\|\Delta\theta_t\|/\|\theta_t\|$, its cosine with $\Delta\theta_{t+1}$ and with the negative gradient $-g_t$ on step t's batch, and the stable and effective rank of its block matrices ΔW. *Curvature*, on 32 fixed training sequences: $\lambda_{\max}(H)$ (20 Lanczos iterations, logged during training), $\operatorname{tr} H$ (10 Rademacher probes), $\operatorname{tr} H/\lambda_{\max}$, and the batch sharpness $\mathbb{E}_B[g_B^\top H_B g_B]/\mathbb{E}_B\|g_B\|^2$. *Predictions*, on the 2,048 held-out positions: entropy, mean top-1 probability, calibration error (15 bins), top-1 accuracy, loss on the targets in the top and the bottom third of training-set frequency and their difference, JS to the unigram and to the bigram model of the training split, and validation minus training loss. *Activations*, on 32 held-out sequences: attention entropy and attention on the first token (means over layers, heads and positions), fraction of MLP pre-activations above 0, RMS of the residual stream after the last block, and effective rank of the centred final features.

Table 4: All metrics of Step 2 on TinyStories-10M (Fig. 2). AdamW, Muon, SGD: mean ± half the range over data orders 0–2, at training loss 1.81. A–M, A–S, M–S: gap between two optimizers on the same data order (mean over data orders) over the larger mean gap between two data orders of one optimizer, median over training losses 1.66–3; positive when the first optimizer is higher. Bold: the five largest in absolute value per column. SGD: heavy-ball momentum 0.9, rate 0.06 (the only rate of 0.03–0.12 that never blew up on three data orders), otherwise as in Table 1.

	AdamW	Muon	SGD	A–M	A–S	M–S
Weights						
weight norm $\|\theta\|_2$	224 ± 2	224.3 ± 0.5	109.5 ± 0.1	−0.3	**51**	**63**
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.155 ± 0.006	0.486 ± 0.006	−1.3	**50**	**58**
largest entry $/\|\theta\|_2$	0.00704 ± 0.00006	0.00645 ± 0.00002	0.0395 ± 0.0010	9.1	−23	−23
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	56.2 ± 5.6	**−147**	−4.4	4.1
effective rank of W	272.2 ± 0.1	324.5 ± 0.1	321.0 ± 0.9	**−263**	−25	2.2
spectral norm of W	5.39 ± 0.07	2.949 ± 0.009	1.47 ± 0.11	74	15	3.0
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.685 ± 0.005	4.37 ± 0.55	**105**	3.7	−1.7
Updates (one step)						
step size $\|\Delta\theta\|/\|\theta\|$	0.00656 ± 0.00008	0.00653 ± 0.00003	0.000624 ± 0.000047	−3.1	**67**	**132**
stable rank of ΔW	16.7 ± 0.3	219.6 ± 0.3	8.37 ± 0.48	**−328**	5.3	**292**
effective rank of ΔW	228 ± 1	369.4 ± 0.2	205 ± 2	**−135**	3.7	46
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	0.857 ± 0.004	25	3.6	−8.8
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	0.447 ± 0.010	2.8	−17	−18
Curvature						
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	50.2 ± 0.6	−10	−11	−18
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	6.38 ± 0.50	−2.7	−5.6	−2.8
Hessian trace	1400 ± 40	2770 ± 20	25500 ± 400	−23	−22	−20
trace / top eigenvalue	224 ± 6	196 ± 7	507 ± 9	−0.5	−7.4	−8.7
Activations						
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	3.84 ± 0.01	−18	−26	−17
attention on first token	0.0175 ± 0.0011	0.0198 ± 0.0005	0.0182 ± 0.0007	−1.3	−1.0	1.2
active MLP units	0.1199 ± 0.0009	0.222 ± 0.001	0.262 ± 0.002	−62	**−56**	−5.5
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	0.244 ± 0.003	28	**78**	**81**
effective rank of features	312 ± 2	319.6 ± 0.2	297 ± 2	−13	−2.7	4.4
Predictions						
prediction entropy (nats)	1.887 ± 0.008	1.878 ± 0.010	1.943 ± 0.005	−0.2	−3.0	−2.5
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.5581 ± 0.0008	−0.4	−0.3	0.3
calibration error	0.0325 ± 0.0017	0.0305 ± 0.0046	0.0297 ± 0.0028	0.1	0.5	0.4
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	0.554 ± 0.003	−1.0	−2.2	−1.2
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	0.851 ± 0.012	0.5	2.5	1.9
loss on rare targets	3.10 ± 0.02	3.102 ± 0.009	3.11 ± 0.04	−0.8	−2.7	−2.1
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	2.26 ± 0.03	−1.1	−3.3	−2.7
JS to unigram model	0.8004 ± 0.0007	0.802 ± 0.001	0.804 ± 0.001	0.8	0.4	−0.7
JS to bigram model	0.4691 ± 0.0010	0.4708 ± 0.0008	0.4622 ± 0.0009	−0.1	2.4	2.3
validation minus training loss	−0.0217 ± 0.0092	−0.0197 ± 0.0062	−0.0224 ± 0.0074	−0.2	0.1	0.3

C Swaps

Table 5: Swap forks on TinyStories-10M. A: parent's optimizer; B: optimizer after the swap; s: fork step.

Parents	the d0 runs of AdamW and Muon (Table 1)
Shadow state	the other optimizer, stepped on the same mini-batch gradients with learning rate 0: momenta, second moments and step counts evolve as if training; parameters follow A alone
Fork steps s	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
Saved at s	parameters; states of both optimizers
Swap $A \to B$ loads	parameters at s; B's shadow state
Hyperparameters	B's (Table 2)
Steps	s to 10,000, on the parent's schedule and batches (data order 0)
Swap forks	2 directions × 9 steps = 18
Same-batch control	the parent itself (A continued with the same batches)
Data-order control	A continued from s with its own state, data order 1: 18 forks

D Compute

All runs used one NVIDIA GPU each on the MIT ORCD cluster. GPU-hours are those Slurm allocated to each suite, shared out over the runs of the suite by their training time where runs of several suites shared a GPU. Not counted: the offline measurements (Muon's spectral-norm sharpness, loss barriers, distances, the metrics of Appendix B), a few GPU-hours.

Table 6: Runs described in the checkpoint and their compute.

	Runs	GPU-hours
Learning-rate sweeps, AdamW and Muon (3 settings)	30	7.1
Runs from the initialization, AdamW and Muon (3 settings, 5 data orders)	30	11.6
Swaps and data-order forks (Table 5)	36	7.1
Step 2: rate search for the added optimizer, runs at the chosen rate	$15 + 3$	3.3
Step 2: runs with consecutive checkpoints (3 optimizers $\times$ 3 data orders)	9	1.9
Total	123	30.9
Running: swaps again, with checkpoints at steps 9998–10,000	18	–